Deliverables: One transducer circuit capable of measuring hot coffee over a suitable range from room temperature to excessively hot, guided by the study cited below. 0.05°C uncertainty.

Ristenpart, W.D., Cotter, A.R. & Guinard, JX. Impact of beverage temperature on consumer preferences*Sci r* black coffee. *Rep* 12, 20621 (2022). https://doi.org/10.1038/s41598-022-23904-4

The measurement error in the results must be $\leq 0.05°C$. We will design with a range of 25°C - ~80°C

In the two articles cited below, the methods used took advantage of multiple thermistor setups.

Guba S, Horváth B, Szalai I. Application and comparison of thermistors and fiber optic temperature sensor referen for ILP measurement of magnetic fluids in double cell magnetic hyperthermia. Heliyon. 2022 Jun 1;8(6):e09606. doi: 10.1016/j.heliyon.2022.e09606. PMID: 35694425; PMCID: PMC9178333.

T. Mathew, C. Anjaneyulu, J. Mohapatra, B. B. R and A. C. Sreekantan, "A Smart Dual-Thermistor-Based Nasal Mask for Respiratory Health Monitoring," in IEEE Sensors Letters, vol. 10, no. 2, pp. 1-4, Feb. 2026, Art no. 2501104, doi: 10.1109/LSENS.2025.3649915

In the first, with a differential thermistor config. in a Wheatstone bridge, the authors were able to achieve a high sensitivity with an amplified differential measurement with two thermistors. Since their application dealt with liquids, this seems like a good direction to proceed in. They also made use of an instrumentation amplifier so we will need to investigate what needs to be amplified and by how much. Their reported sensitivity was high as well, at $-72.3\ mV/C$. They did not acheive an accuracy like our target 0.05°C, so we will need to figure out if the differential Wheatstone thermistor can achieve this accuracy, if we want to retain the sensitivity characteristic.

Previously, we found a $0.70 thermistor called 103AP-2A by Semitec. While it boasts of a ±0.5 resistance tolerance, it is the ±0.5 β tolerance that most intrigues me, having previously seen the ± 0.75 β tolerance of the NTCLE100E3206B0. With a lower β tolerance, we can more accurately control error at those higher temps.

GOAL 1

The next goal is to rerun the existing jupyter notebook for the 103AP's tolerance characteristics. An Excel sheet will be started with design decisions relevant to source voltage decisions and the noise thereof, and an amplifier if needed.

LTspice Modeling:

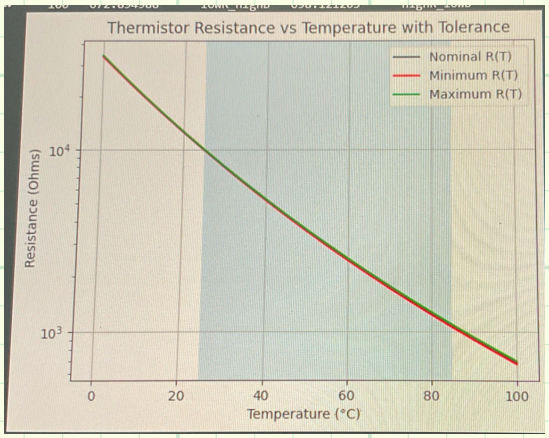
Could not find official model for it. options are to model a behavioral resistor according to LTspice forums. GOAL 2

Considering T_null: GOAL 3

Considering Self-Heating GOAL 4

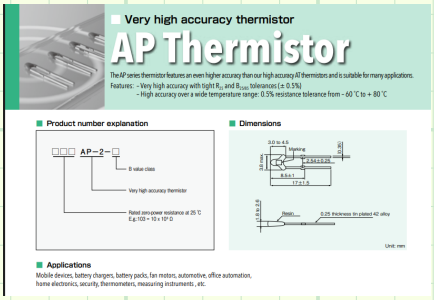
Using Thermistor 1 as Ref GOAL 5

153 results, cheapest option is 103AP-2-A at $0.70 USD, ±0.5% R_{25} tolerance and a ±0.5% $B_{25/85}$ tolerance



Even with a lower tolerance, the lowest R_T was higher than the previous.

For NTCLE:	Min Ω	Max Ω	Nom. Ω	Min %	Max %
25°C	9800	10200	10,000	98	1.02
50°C	3404.948	3662.63	3563.13	97.24	1.028
For 103AP:					
25°C	9950	10050	10,000	99.5	1.005
50°C	3527.99	3600.401	3564.05666	98.9	1.0101



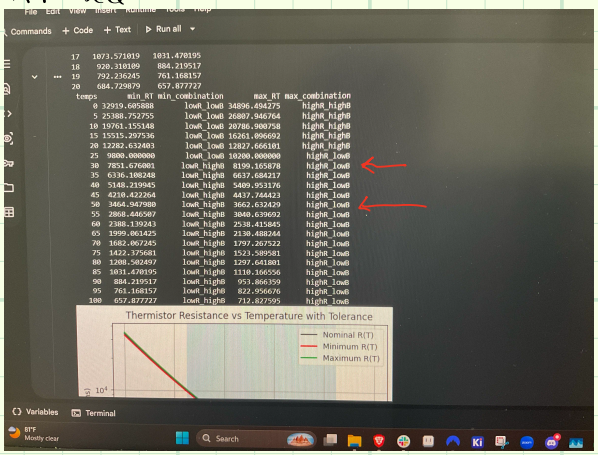
Specifications

Product number	R_{25}[1]	R_{25} tolerance	B value[2]	Dissipation factor (mw / °C)	Thermal time constant (s)[3]	Rated power at 25 °C (mW)	Operating temperature range (°C)
202AP-2	2.00 kΩ	± 0.5%	3976 K ± 0.5%	approx. 1.2	approx. 15	6	– 60 to 150
232AP-2	2.252 kΩ		3976 K ± 0.5%				
502AP-2	5.00 kΩ		3976 K ± 0.5%				
103AP-2	10.0 kΩ		3435 K ± 0.5%				
103AP-2-A			3976 K ± 0.5%				
203AP-2	20.0 kΩ		3976 K ± 0.5%				
503AP-2	50.0 kΩ		4220 K ± 0.5%				
104AP-2	100 kΩ		4261 K ± 0.5%				
204AP-2	200 kΩ		4470 K ± 0.5%				

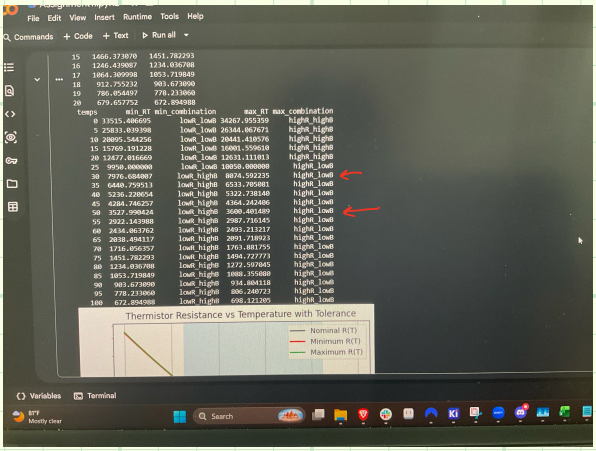
[1]: Rated zero-power resistance at 25 °C [2]: B value calculated from rated zero-power resistance at 25 °C and 85 °C
[3]: Time required to reach 63.2% of temperature difference. Measured with sensor suspended in mid-air.

https://www.semitec-global.com/uploads/2022/01/P10-11-AP-Thermistor.pdf

NTCLE:



103AP:



How low should the self-heating effect be to minimize effects?

Self-Heating due to a Constant Current

$$P = I^2 R_T = \Delta T_{sh} \delta$$

$$\Delta T_{sh} = \frac{P}{\delta}$$

$$\Delta T_{sh} = \frac{I^2 R_T}{\delta}$$

No compensation in this eq for resistance change.

Good enough for small changes

E.g.

$$P = I^2 R_T$$

$$P = (100\mu A)^2 (10000\Omega) = 100\mu W$$

$$\Delta T = \frac{P}{\delta} = \frac{100\,\mu W}{7 mW/K} = 0.0143°C$$

7 mW/K from datasheet

$$\Delta R_T = \frac{dR_T}{dT} \Delta T_{sh} = -\frac{\beta}{T^2} R_T \Delta T_{sh}$$

$$\Delta R_T = (-447\Omega/K)(0.0143K) = -6.4\Omega$$

due to self-heating

Impact of Excitation Current Magnitude

how large before self-heating stops being negligible against 0.05°C change

Looking ahead at thermistors:

Ex.

$V_S = 3.3V \quad R_1 = R_2 = R_S = 10K\Omega$

Current division: $I_T = \frac{V_S}{R_S + R_T}$

$$I_T = \frac{3.3}{10e3 + 10e3} = 165\mu A$$

Calculate Power: $P_T = I_T^2 R_T$

$$P_T = (165\mu A)^2 (10000\Omega)$$

$$P_T = 272\mu W$$

Self-heating: $\frac{165\,\mu A}{272\,\mu W} = 0.039°C$ and a resistance change of -17.4Ω, which is 78% of the 0.05°C needed